MUSE User's Manual

Increase the knowledge of Natural Systems through the evaluation of the uncertainty of environmental data: operational theory and application

1 Introduction

The Ph.D. Project addresses issues in the management and processing of geospatial information for modeling Natural Systems, paying particular attention to the "operational" value of data and uncertainty evaluation.

To fully understand a Natural System, representing an environmental variable's distribution in 3D space is a mandatory and complex task. The challenge derives from a scarcity of samples number in the survey domain (e.g., logs in a reservoir, soil samples, fixed acquisition sampling stations) or an implicit difficulty in the in-situ measurement of parameters. Field or lab measurements are generally considered error-free, although not so. This aspect, combined with conceptual and numerical approximations used to model phenomena, makes the results less performing, fading the interpretation [?]

In this context, a computational infrastructure is proposed to evaluate the spatial uncertainty in a multi-scenario application in Environment survey and protection. In the geoscience field, different software exists to model geodata and perform simulations. They are distinguished according to their application sector; among the open-source solutions, different categories exist, such as seismic and seismology, simulation and modeling (e.g., *MODFLOW* [?]), reservoir engineering, geostatistics (e.g., *GSLIB* [?], *gstat* [?], *SGeMS* [?], ...), geospatial management (e.g., GRASS GIS [?]), geochemistry (e.g., *PHREEQC* [?]), just to mention the most important. In addition to being related to specific application sectors, these tools are developed in accordance with the data, and have limitations in terms of input parameters or the algorithms list.

The research interest becomes to expand the operative knowledge by developing an open-source stochastic tool, named MUSE (*Modeling Uncertainty as a Support for Environment*), based on a solid theoretical-based methodology and with a general and non-sectoral footprint. It is conceived to work indiscriminately in various application fields, from environmental geochemistry to coastal oceanography, to infrastructure engineering, or in any scenario in which the estimate of the distribution of a variable measured on the survey volume assumes relevance. The decision to focus on open-source software development is made for the possibility of analyzing specific algorithms to produce expected results. The open-source feature is very interesting for using the tool in scientific research as its functionalities are more accessible.

In MUSE, the innovative contribution is to give a different role to uncertainty. Starting from the definition of regionalized variables, the uncertainty will be quantified and used for more conscious approaches to decisions. The goal will be to consider not only the uncertainty on the results but also on input data, widening the analysis to ranges or measurements with their confidence interval (i.e., *soft data*) as input. By that approach uncertain data, derived also from expert opinion and a-priori statistical knowledge, will help users to understand spatial variability.

The developed tool will be addressed to environmental researchers, Environmental Agencies, Research Institutions, and policy- or decision-makers. Additionally, the research results must be consistent with the development criteria required by a Technology Readiness Level TRL 5-9[1].

2 Development criteria

To deal with uncertainty, it is necessary to design a robust tool capable of managing heterogeneous environmental data and geospatial information in a flexible way and performing stochastic analysis. Therefore, the research activities are focused on the definition and development of a stable structure of the computational code to process geodata in specified computational domains. During the code development, the following criteria are chosen to achieve the intended purposes:

1. *Generalized approach for input data, and on spatial discretization*
   The variability of environmental data types is addressed by implementing a general method to encode heterogeneous data, and consequently by adopting a specialized data format (see Section **??**). Standard methods in computational geometry are applied for localization and spatial data management, adapting the state-of-the-art algorithms to the needs of the application. Spatial discretization exploits the concept of polygon mesh in two-dimension, and polyhedral mesh in

[1] TRL 5: Technology validated in relevant environment (industrially relevant environment in the case of key enabling technologies); TRL 9: Actual system proven in operational environment (competitive manufacturing in the case of key enabling technologies; or in space).

three-dimension (as a collection of nodes, linked by edges). Polygons and polyhedral can assume any shape. By this geological bodies can be described adaptively in a much more detailed manner. At this stage, the code supports triangular/tetrahedral meshes (see Section **??**); as needed, the geometric module will be extended to support also regular grids, and volumetric counterparts (i.e., voxels).

2. *Stratigraphic transformation in 2D/3D space*
   COMPLETARE

3. *Software design: additivity, geo-metadata system and specifications*
   From the point of view of software design, the main goal is to create an open-source tool and a modularized procedure to make the tool as robust as possible, and easily upgradeable with a wide set of algorithms, even at a later stage. Each single calculation step (see Section **??**) is featured by information storage in a series of geo-metadata files, to be used for the ensuing steps. This geo-metadata system permits to increase the knowledge of the final model and to trace back the processing carried out on the input, assuring reproducibility of the model at any time.

4. *Geospatial analysis following EPSG Standard*
   A powerful feature available to all geomodeling and computing modules enables the usage of environmental data based on whatsoever coordinate reference system. The management of the EPSG code is guaranteed by using PROJ library [**?**], as an external dependence of the `MUSE` system. This part is under development to enable geodesic computation over any reference datum.

5. *Availability of a stochastic computational code*
   At the current status, the module of geostatistical computing includes *Sequential Gaussian Simulations* algorithm for continuous variables (see Section **??**) as result of an INTERREG Project. The implementation for discrete variables is foreseen, and further computational codes (i.e., stochastic as *simulated annealing* and *cosimulations* or deterministic as *IDW*, *splines* or *Radial Basis Functions*) are planned for the second year of activity. This will increase the available algorithms in the library, improving the computational functionalities of `MUSE`.

6. *Domain's segmentation*
   To better manage the geological or more generally environmental spatial heterogeneity of variables distributions, the analysis can be performed on independent sub-domains. Sub-domains are recognized automatically and extracted from geometric model. A tool to extract data, located in different 2D and/or 3D sub-domains, is implemented to properly compute the covariance law (by variography) and simulations.

7. *Data processing*
   Analysis of static dataset, so for ex-situ processing, were privileged as the first step to consolidate the methodology. Then, a real-time module will be developed for in-situ data processing.

3 MUSE Modeling Uncertainty as a Support for Environment Software

MUSE is an open-source software package.

It is developed for ...

It includes ... modules.

MUSE is an embedded package (all useful functionalities are self-contained): for this reason, the process from input to output is automatic and implicit, without further coding complications. Combining state-of-the-art techniques and advanced algorithms (e.g., parallel computing) guarantees robustness and, at the same time, high computing performances. The code is developed in C++ programming language and will be available for different Operating Systems (Linux, Mac, and Windows OS).

MUSE is used by flexible and concise language in a command-line interface. The source code repository contains the entire MUSE package, an examples folder with useful sample data, and the documentation. Referring to examples, data distributed with the source code are related to real case studies or artificially created for analyzing specific interesting situations. The proposed examples are discussed in Section **??**. The documentation, as a list of command-line arguments, is illustrated in Appendix **??**.

3.1 MUSE modules

... ...

4 Documentation

In the following sections, a technical manual is proposed for helping to use MUSE applications.

4.1 Data

MUSE Data is used for loading and managing heterogeneous environmental data through the implementation of a flexible descriptive format. The functionalities related to CSV reading, checks on variables, conversion in MUSE Data Format, and preliminary statistics on input are fully functional for ex-situ analysis. Usage:

- -N, --new_project — Creation of a new project
- -p <path>, --pdir <path> — (Required) Project directory
- --setEPSG <authority> — Set Project EPSG
- -S, --setIDXYZ — Set n. column coordinate
- --setID — Set ID
- --setX — Set X coordinate
- --setY — Set Y coordinate
- --setZ — Set Z coordinate
- --setDel <DEFAULT | COMMA> — Set type of CSV delimiter (default = DEFAULT[2])
- -C, --converter — Converter data (CSV format) into MUSE format
- --inf <inf> — Set inf limit for bounded values (default = 0)
- --sup <sup> — Set sup limit for bounded values (default = 1)
- -R, --read — Reading MUSE Data Format
- --nvalhist <int> — Set minimum number of values for histogram (default = 20)

- --, --ignore_rest — Ignores the rest of the labeled arguments following this flag.
- --version — Displays version information and exits.
- -h, --help — Displays usage information and exits.

[2] DEFAULT corresponds to semi column delimiter.

4.2 Geometry

MUSE Geometry is the geometry processing module for 2D and/or 3D discretization of the space for the reconstruction of computing domains, after loading of available geospatial information. Outputs are saved in standard formats for polygonal and polyhedral meshes. The default for polygonal mesh is *.off*, and for polyhedral mesh is *.mesh*. Usage:

-N, --new_project	Creation of a new project
-p <path>, --pdir <path>	(Required) Project directory
--setEPSG <authority>	Set Project EPSG
-V, --vector	Load Vector file (.shp, .gpkg)
-R, --raster	Load Raster file (.ASCII)
-P, --pcloud	Load point cloud (.txt, .dat)
--points <filename>	Load filename as POINTS geometry type
--polygon <filename>	Load filename as POLYGON geometry type
--rotaxis <rot_axis>	Set rotation axis
--rotangle <double>	Set clockwise rotation angle (in degree)
--rotcx <double>	Set coordinate X of the rotation center
--rotcy <double>	Set coordinate Y of the rotation center
--rotcz <double>	Set coordinate Z of the rotation center
--tri	Set triangulation for 2D meshing
--convex	Set convex hull for points triangulation
--concave	Set concave hull for points triangulation
--boundary <filename>	Load external boundary for points triangulation
--opt	Set optimization parameters for triangulation (refers to [?] documentation)
--grid	Set grid for 2D meshing
--resx <double>	Set resolution in X direction (default = 1.0)
--resy <double>	Set resolution in Y direction (default = 1.0)
--resz <double>	Set resolution in Z direction (default = 1.0)
-O, --offset	Apply offset on 2D mesh
--delta	Set DELTA offset
--abs	Set ABSOLUTE ELEVATION offset
-z <double>, --zoffset <double>	Set offset in Z direction
-A, --append	Append meshes
-T, --triobj	Create an object closed by two triangular meshes
-Q, --quadobj	Create an object closed by two quadrilateral meshes
-m <filename>, --mesh <filename>	(Accepted multiple time) Set meshes to load
-M, --volmesh	Create polyedral mesh from loading polygonal mesh

`--tet`	Set tetrahedralization for 3D meshing
`--vox`	Set voxelization for 3D meshing
`--nmaxvox <int>`	Set maximum number of voxel per side (default = 1)
`--hex`	Set hexahedralization for 3D meshing
`--obj`	Saving 2D mesh in OBJ format
`--vtk`	Saving 3D mesh in VTK format
`-L, --trimesh`	Load triangular mesh
`--newres <string>`	Set new resolution
`--splmet <string>`	Set polys split method
`--, --ignore_rest`	Ignores the rest of the labeled arguments following this flag.
`--version`	Displays version information and exits.
`-h, --help`	Displays usage information and exits.

4.3 Manipulate

`MUSE Manipulate` extracts input data from independent domains, exploiting algorithms for geometry processing in 2D and/or 3D space. The application also includes functionalities to transform the model in the stratigraphic coordinate system, to better consider the spatial correlation of samples in specific geological scenarios. Usage:

`-E, --extract`	Sampled data extraction from specific geometry model
`-p <path>, --pdir <path>`	(Required) Project directory
`--geom <filename>`	Set geometry model
`--rotaxis <rot_axis>`	Set rotation axis
`--rotangle <double>`	Set clockwise rotation angle (in degree)
`--rotcx <double>`	Set coordinate X of the rotation center
`--rotcy <double>`	Set coordinate Y of the rotation center
`--rotcz <double>`	Set coordinate Z of the rotation center
`-S, --prsect`	Compute points projection on boundary (2D section case)
`-V, --prvol`	Compute points projection on boundary (3D volumetric case)
`--prdir <string>`	Set axis direction of projection (default = Y)
`--sub <string>`	Set extracted sub-dataset referring to specified geometry domain
`--type <SAMPLES | GEOMETRY | VOLUME>`	Set input type for computing points projection
`-T, --strat`	Compute stratigraphic coordinate transformation
`--sttype <PROPORTIONAL | TRUNCATION | ONLAP | COMBINATION>`	Set stratigraphic condition for coordinate transformation
`--top <string>`	Set name of top geometry model
`--bot <string>`	Set name of bottom geometry model
`--obj`	Saving 2D mesh in OBJ format
`--vtk`	Saving 3D mesh in VTK format
`--, --ignore_rest`	Ignores the rest of the labeled arguments following this flag.
`--version`	Displays version information and exits.
`-h, --help`	Displays usage information and exits.

4.4 Vario

MUSE Vario is able to calculate 3D omnidirectional, and 2D directional variograms, for continuous and discrete variables, in both automatic and manual modes. Usage:

-V, --variogram	Compute variogram
-p <path>, --pdir <path>	(Required) Project directory
-v <name>, --var <name>	(Required) Variable
-z <zvar>, --setZ <zvar>	Set Z coordinate
--sub <string>	Set extracted sub-dataset referring to specified geometry domain
--rotaxis <rot_axis>	Set rotation axis
--rotangle <double>	Set clockwise rotation angle (in degree)
--rotcx <double>	Set coordinate X of the rotation center
--rotcy <double>	Set coordinate Y of the rotation center
--rotcz <double>	Set coordinate Z of the rotation center
--nscore <string>	Set Normal Score transformation (default = NO)
--decl <string>	Set 2D declustering (deafult = NO)
--csize <double>	Set 2D cell size for 2D declustering
--nstep <int>	Set n steps for 2D declustering
--sttype <PROPORTIONAL | TRUNCATION | ONLAP | COMBINATION>	Set stratigraphic condition for coordinate transformation
--vario <EXPERIMENTAL | MODEL>	Set type of variogram (default = EXPERIMENTAL)
--dir <OMNI | DIR>	Set variogram direction (default = OMNI)
--dim <3D | 2D | 1D>	Set variogram dimension (default = 3D)
--lagspac <VARIABLE | CONSTANT>	Set lag spacing type (default = VARIABLE)
--deg <degree>	Set degree step in counter-clockwise from North (in degree) (default = 45°)
--degtol <degree>	Set degree tolerance (in degree) (default = 45°)
--dirs <dir0,dir1,...,dirn>	Load n discrete directions (in degree) for computing directional variogram
--type <SPHERICAL | GAUSSIAN | EXPONENTIAL >	Set model type
--, --ignore_rest	Ignore the rest of the labeled arguments following this flag.
--version	Displays version information and exits.
-h, --help	Displays usage information and exits.

4.5 Compute

`MUSE Compute` is the geostatistical processing module for the evaluation of the spatial heterogeneity of the sampled variables, supported by specific covariance laws provided by `MUSE Vario`. At the current status, it works on continuous variables, exploiting *Sequential Gaussian Simulation* (SGS) algorithm.

mari: Con il –mode AUTO, compute proceed con il calcolo classic delle simulazioni gaussiane: ovvero considers tutti I punti campionati ed il variogramma ad essi associate. Per analisi più rigorose, in cui il numero di punti risulta insufficient al calcolo del variogramma, una ulterior modalità di calcolo è state aggiunta per consentire di considerare un variogramma calcolato su un numero di punti diverse rispetto ai campioni reali (di numero ad esempio inferior) - vedi esempio Chiesa di Santa Margherita. Usage:

`-C, --compute`	Start Sequential Gaussian Simulations
`-p <path>, --pdir <path>`	(Required) Project directory
`-v <name>, --var <name>`	(Required) Variable
`-S, --stats`	Start Sequential Gaussian Simulations
`-B, --backnscore`	Set Back Normal Score
`--, --ignore_rest`	Ignores the rest of the labeled arguments following this flag.
`--version`	Displays version information and exits.
`-h, --help`	Displays usage information and exits.

4.6 Realtime

MUSE Realtime Usage:

-N, --new_project	Creation of a new project
--setEPSG <authority>	Set Project EPSG
--, --ignore_rest	Ignores the rest of the labeled arguments following this flag.
--version	Displays version information and exits.
-h, --help	Displays usage information and exits.